

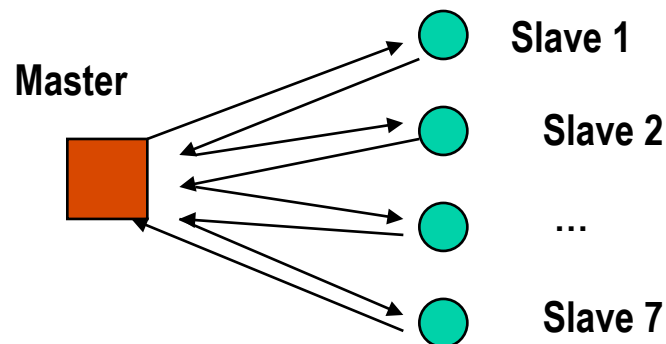
Ex. 1a

Bluetooth uses a Polling mechanism between the *Master* and the *Slaves* to allow communication.

Compute the efficiency of this *Roll-Call Polling* mechanism, when the total number of *Slaves* is equal to 7, but only 2 of them always have packets to transmit.

Packets have a size equal to 2744 bit, and the *token* is of 100 bit. The propagation delay between each slave station and the Master is equal to $10\ \mu\text{s}$, the channel capacity is 723 kbit/s.

Bluetooth - Roll-Call Polling



Ex 1a - Solution

N=2 active *Slaves*

M=7 total *Slaves*

$$T_P = \frac{2744}{723000} = 3.795ms$$

$$T_T = \frac{100}{723000} = 138.3\mu s$$

$$\tau = 10\mu s$$

Useful time (effectively used to transmit data) = NT_P

Total duration of a cycle:

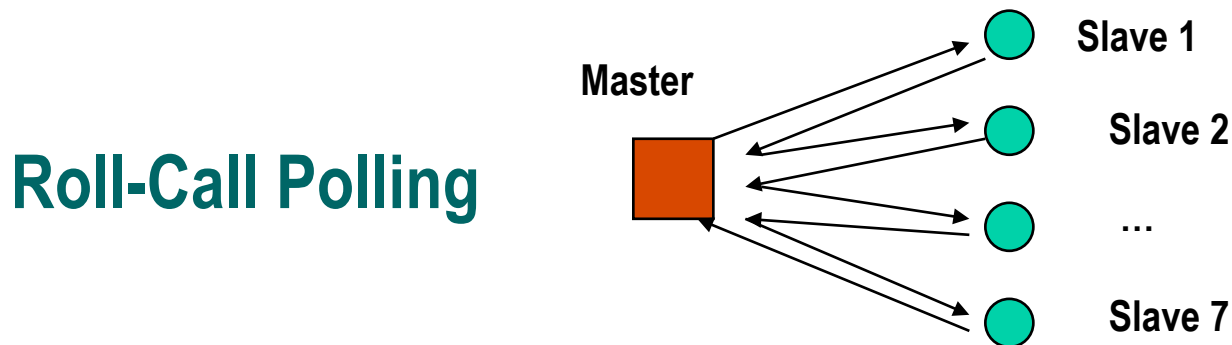
$$(M - N)(2\tau + 2T_T) + N(2\tau + 2T_T + T_P) = 2M(\tau + T_T) + NT_P$$

$$\text{Efficiency : } \eta = \frac{NT_P}{2M(\tau + T_T) + NT_P} = 0.785 = 78.5\%$$

Ex. 1b

In the same context of Ex. 1a, compute the efficiency of the *Roll-Call Polling* protocol of Bluetooth, in case the 2 active *Slaves* transmit packets of size equal to **200** bit.

All other parameters are the same: the *token* is 100 bit long. The propagation delay between each slave station and the Master is equal to $10\ \mu\text{s}$, the channel capacity to 723 kbit/s.



Ex 1b - Solution

N=2 active *Slaves*

M=7 total *Slaves*

$$T_P = \frac{200}{723000} = 0.2766ms$$

$$T_T = \frac{100}{723000} = 138.3\mu s$$

$$\tau = 10\mu s$$

Useful time (effectively used to transmit data) = NT_P

Total duration of a cycle:

$$(M - N)(2\tau + 2T_T) + N(2\tau + 2T_T + T_P) = 2M(\tau + T_T) + NT_P$$

$$\text{Efficiency } \eta = \frac{NT_P}{2M(\tau + T_T) + NT_P} = 0.2104 = 21.04\%$$

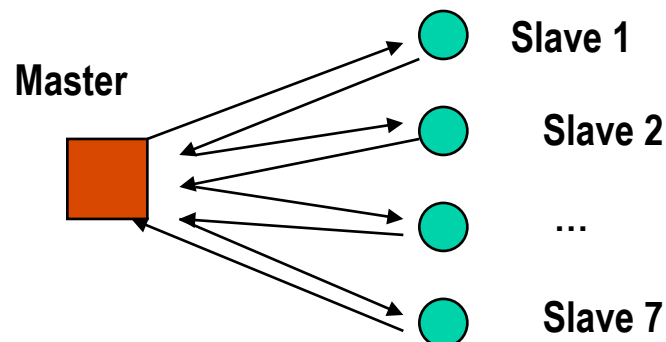
Ex. 1c

Compute the maximum time necessary for a *Slave* (for example, *Slave 1*) to access the channel in order to transmit a packet to the *Master*.

The total number of *Slaves* is always equal to 7, and let us suppose that all stations have packets to transmit.

Packets have a size equal to 2744 bit, the token is 100 bit. The propagation delay between each station and the Master is $10\text{ }\mu\text{s}$, the channel capacity is 723 kbit/s.

Roll-Call Polling



Ex 1c - Solution

N=7 active *Slaves*

M=7 total *Slaves*

$$T_P = \frac{2744}{723000} = 3.795ms$$

$$T_T = \frac{100}{723000} = 138.3\mu s$$

$$\tau = 10\mu s$$

Time necessary for the M-1=6 other stations to transmit their packets:

$$(M - 1)(2\tau + 2T_T + T_P) = 24.55ms$$

Finally, to be precise, it is necessary for Slave 1 to send its token to the Master and receive it again:

$$T_{\text{access}} = 24.55ms + 2(\tau + T_T) = 24.85ms$$

Ex. 1d

Let us imagine now that, in Bluetooth, the *Master* uses a Polling mechanism towards all stations, including not only the *Slaves* but also *Parked Slaves*.

Compute the efficiency of the *Roll-Call Polling* mechanism, in case the total number of Slaves is equal to 7 (and 2 of them have packets to send, as in Exercise 1a), and the number of *Parked Slaves* is equal to 256 (by definition, *Parked Slaves* do not have packets to transmit).

Packets have a size equal to 2744 bit, and the *token* is of 100 bit. The propagation delay between each station and the Master is equal to 10 μ s, the channel capacity to 723 kbit/s.

Ex. 1d - Solution

N=2 active slaves

M=7+256=263 total slaves

$$T_P = \frac{2744}{723000} = 3.795ms$$

$$T_T = \frac{100}{723000} = 138.3\mu s$$

$$\tau = 10\mu s$$

Useful time (effectively used to transmit data) = NT_P

Total duration of a cycle:

$$(M - N)(2\tau + 2T_T) + N(2\tau + 2T_T + T_P) = 2M(\tau + T_T) + NT_P$$

$$\text{Efficiency: } \eta = \frac{NT_P}{2M(\tau + T_T) + NT_P} = 0.08867 = 8.867\%$$